

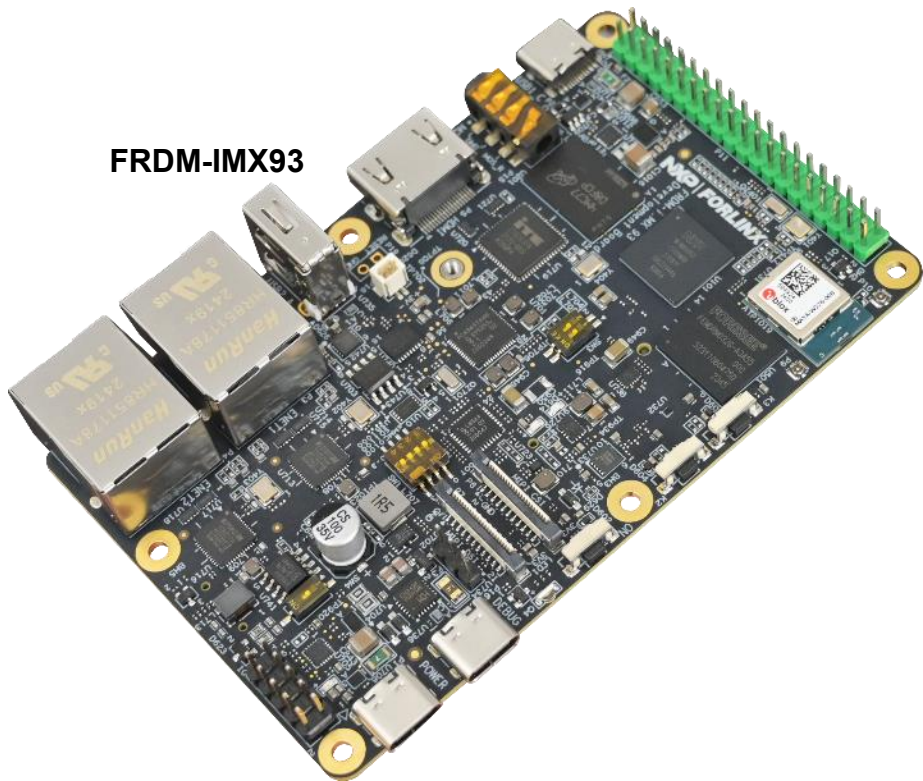
MakeNTU NXP FRDM-IMX93 Introduction

Mar.2026



/ Agenda

FRDM-IMX93



HW overview

- i.MX 93 introduction
- FRDM-IMX93 Features

SW environment

- Flash image
- Build Embedded Linux

Working with accessories

- Use USB CAM
- Enable WiFi
- Basic IO control

GoPoint / NPU

HW overview

/i.MX93 - Block Diagram



i.MX 93 Applications Processors

Multicore Processing

- Arm® Cortex®-A55 @ 1.7 GHz (x2)
- Arm® Cortex®-M33 @ 250Mhz
- Arm® Ethos™ U-65 microNPU
- EdgeLock® secure enclave

External Memory

- Up to 3.7GT/s x16 LPDDR4/LPDDR4X
- 3x SD 3.0/ SDIO3.0/ eMMC5.1
- Octal SPI, including support for SPI NOR and SPI NAND memories

Graphics

- Hardware Compositor for Blending/Composition, Resize, Color Space Conversion

Display Interfaces

- 1080p60 MIPI-DSI (4-lane, 1.5Gbps/lane)
- 720p60 LVDS (4-lane)
- 24-bit parallel RGB

Camera Interfaces

- 1080p60 MIPI-CSI (2-lane, 1.5Gbps/lane)
- 8-bit parallel YUV/RGB

Audio Interfaces

- 7x I2S TDM (32-bit @ 768KHz), SPDIF Tx/Rx
- 8 channel PDM microphone input
- MQS: Medium Quality Sound output

Operating Systems

- Linux® OS, FreeRTOS, QNX

/ Get to know **FRDM i.MX 93 Board**

2x **Ethernet** ports

MIPI-CSI, MIPI-DSI and HDMI
connectors



2x20 header for
expansion boards
and **adapters**

Debug, PD and USB type-C
+ **USB** Type-A

IW612 Wireless Module - **u-**
blox: MAYA-W276

- **TRI-RADIO** Wi-Fi 6 + Bluetooth + 802.15.4
- NXP EdgeLock® Security
- Matter over Wi-Fi & Thread
- Thread Border Router
- Thread or Zigbee
- BDR, EDR and Bluetooth LE

Specifications

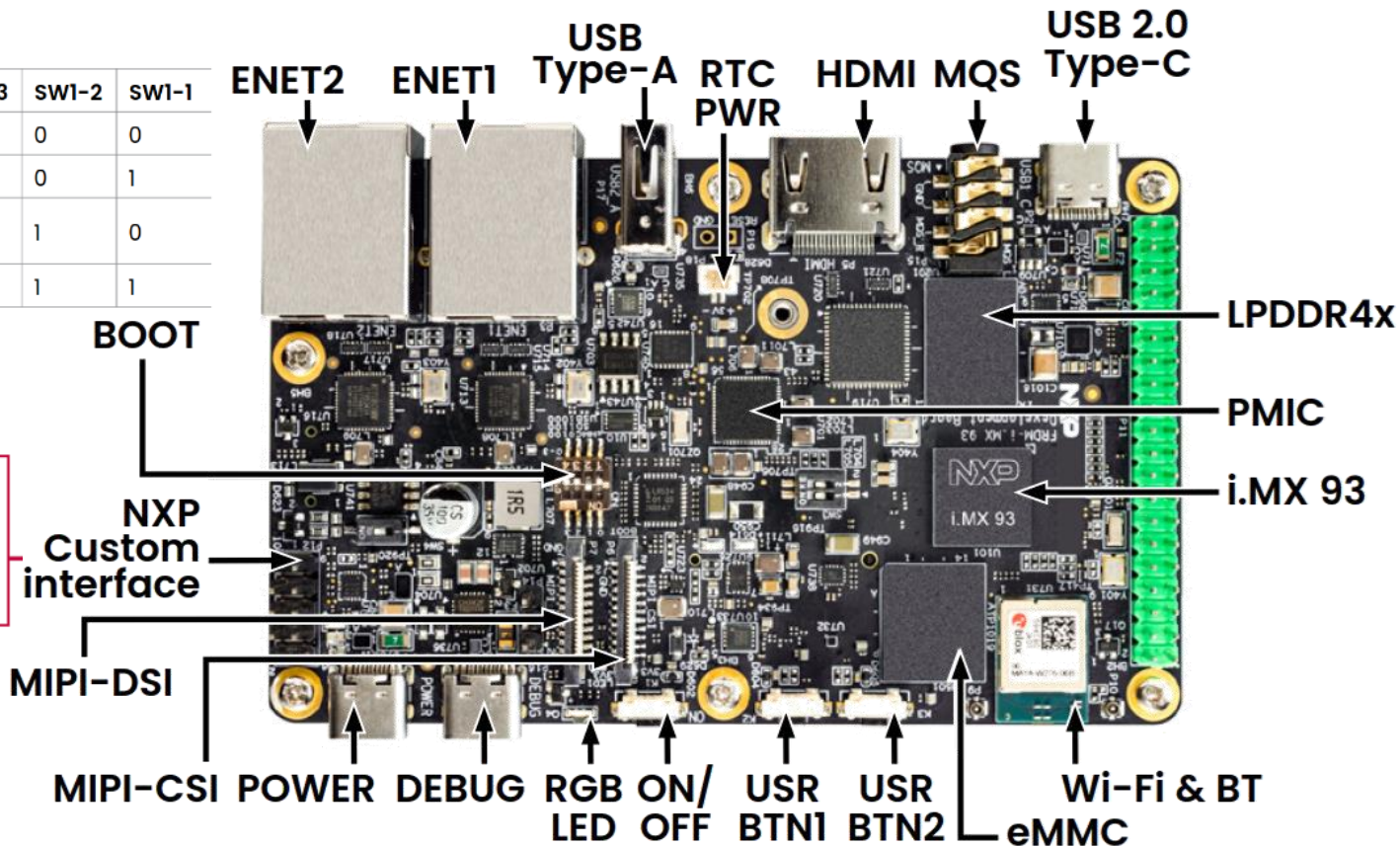
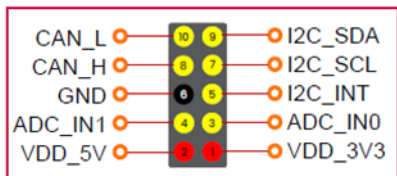
- 2x Arm Cortex®-A55 + Cortex®-M33
- Wi-Fi 6 + BT + 802.15.4 Module on-board, IW612
- 2x GB Ethernet (1xETER, 1xTSN)
- RPI MIPI-CSI/DSI, HDMI
- M.2 Connector
- LPDDR4X 16-bit 2GB
- eMMC 5.1, 32GB
- MicroSD 3.0 card slot
- 3x USB 2.0 Type-C connector (one for Debug, one PD only) + 1x USB 2.0 Type-A
- RTC, Buttons and LED

NXP Devices On-Board

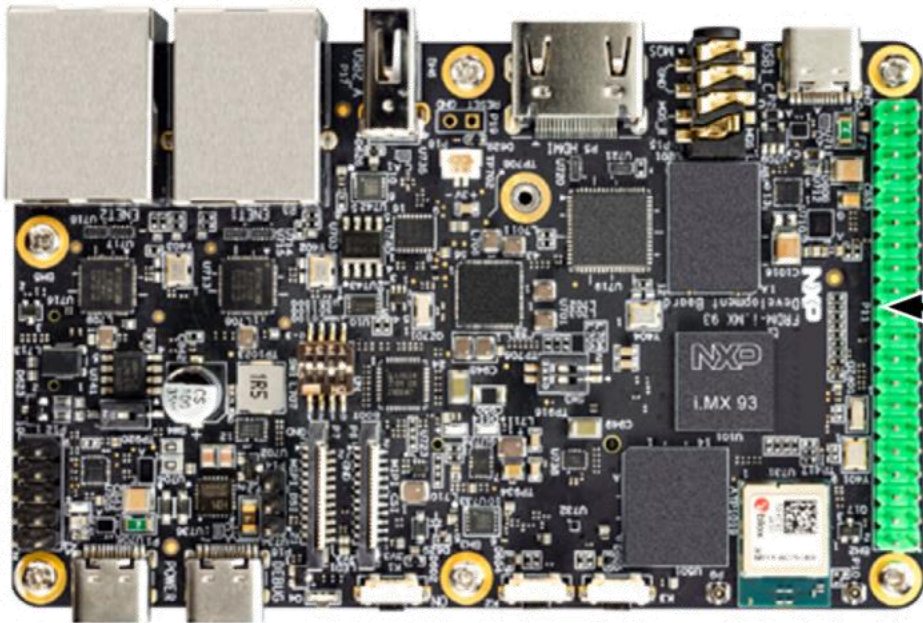
PMIC	PCA9451A
USB PD TCPC PHY IC	PTN5110
High-Voltage USB PD Power Switch	NX20P5090UK
IIC Extends GPIO	PCAL6524/PCAL6408A
CAN Transceiver	TJA1051
USB Sink & Source combo power switch	NX20P3483UK
USB Type-C CC and SBU Protection IC	NX20P0407
Real-time clock/calendar	PCF2131
Wi-Fi, BT, 802.15.4 Tri-Radio	IW612 (in u-blox Module)

FRDM i.MX93 Board - Front

BOOT mode	SW1-4	SW1-3	SW1-2	SW1-1
From internal fuses	0	0	0	0
Serial downloader	0	0	0	1
USDHC1 8-bit eMMC 5.1	0	0	1	0
USDHC2 4-bit SD3.0	0	0	1	1



/FRDM i.MX93 Board - Front



	3V3	1	2	5V	
I2C4.SDA	GPIO_2	3	4	5V	
I2C4.SCL	GPIO_3	5	6	GND	
PDM.CLK	GPIO_4	7	8	GPIO_14	UART3.TXD
	GND	9	10	GPIO_15	UART3.RXD
	GPIO_17	11	12	GPIO_18	SAI3.RXBCLK
	GPIO_27	13	14	GND	
	GPIO_22	15	16	GPIO_23	
	3V3	17	18	GPIO_24	
SPI3.SOUT	GPIO_10	19	20	GND	
SPI3.SIN	GPIO_9	21	22	GPIO_25	
SPI3.SCK	GPIO_11	23	24	GPIO_8	SPI3.CS0
	GND	25	26	GPIO_7	SPI3.CS1
	GPIO_0	27	28	GPIO_1	
PDM.BITSTREAM_0	GPIO_5	29	30	GND	
PDM.BITSTREAM_1	GPIO_6	31	32	GPIO_12	PDM.BITSTREAM_2
PDM.BITSTREAM_3	GPIO_13	33	34	GND	
SAI3_RXSYNC	GPIO_19	35	36	GPIO_16	SAI3.TXBCLK
SPI3.TXSYNC	GPIO_26	37	38	GPIO_20	SAI3.RXDATA0
	GND	39	40	GPIO_21	SAI3.TXDATA0

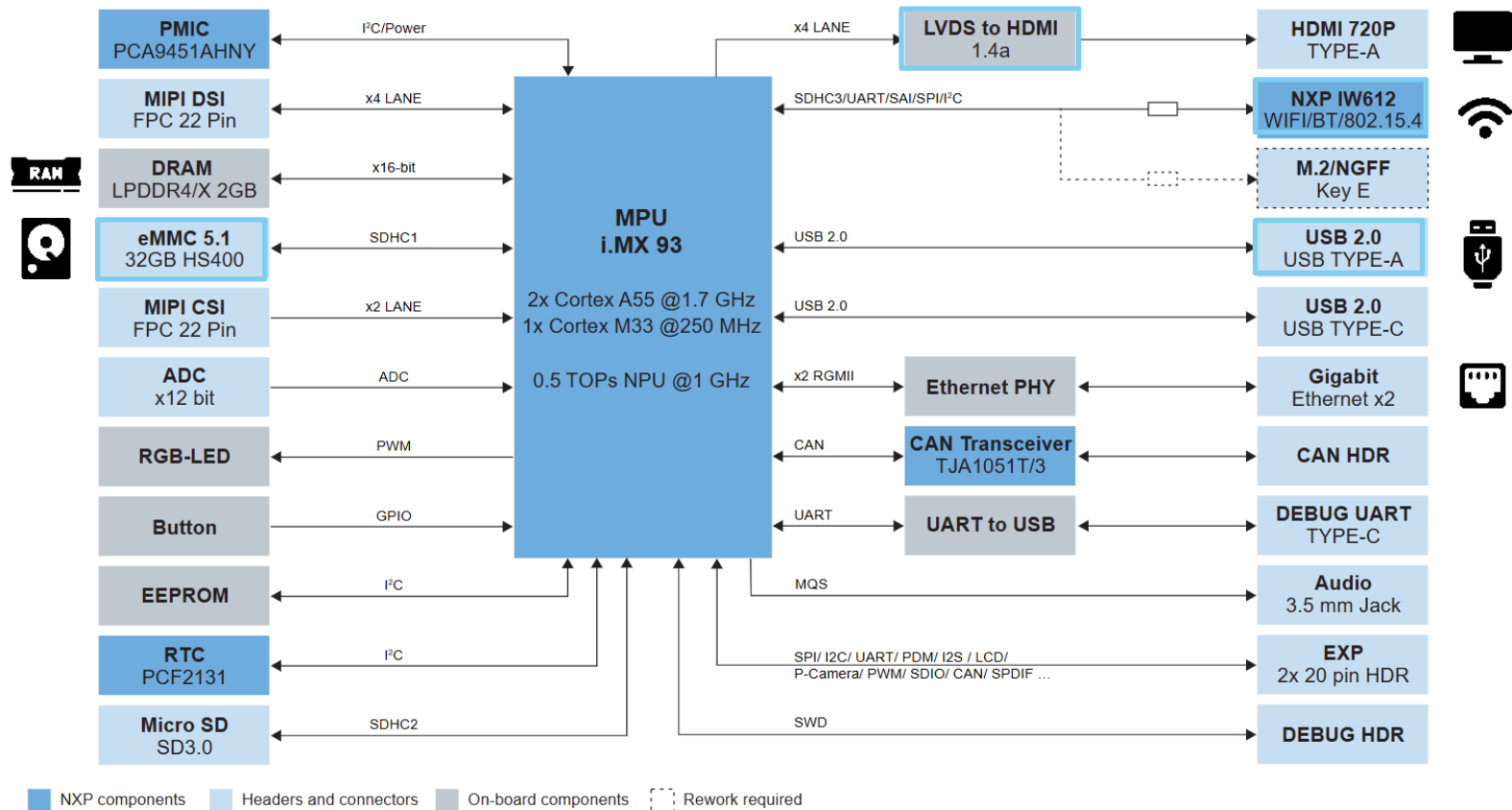
/FRDM i.MX93 Board - Back



M.2 slot

MicroSD slot

FRDM i.MX93 Board - Block Diagram



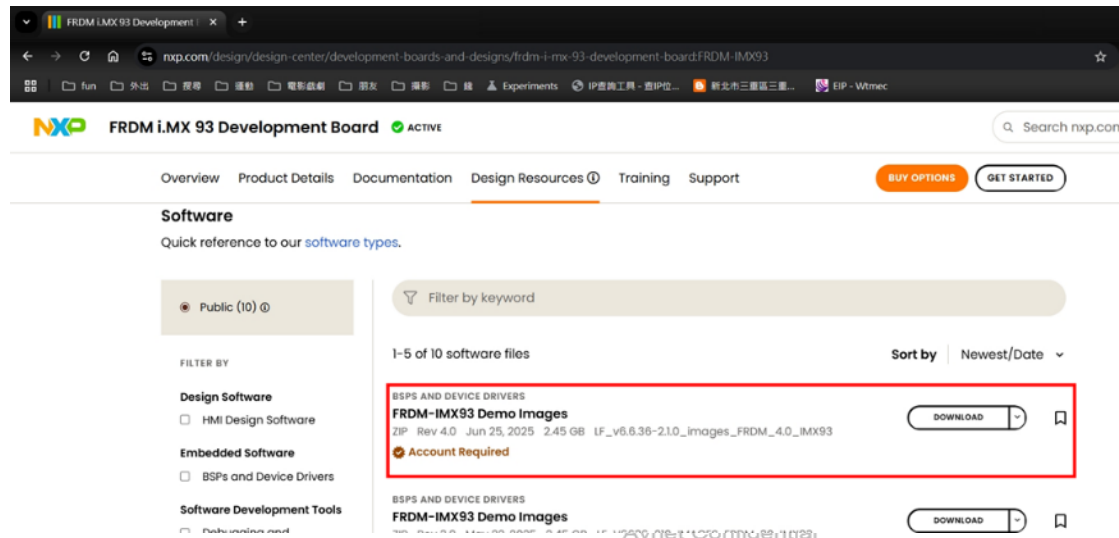
SW environment

/ Getting Started with FRDM i.MX93

Step 1

Download Pre-Build Image (Yocto Linux)

<https://www.nxp.com/design/design-center/development-boards-and-designs/frdm-i-mx-93-development-board:FRDM-IMX93>



/Getting Started with FRDM i.MX93

Step 2

1. Using uuu to burn image and rootfs to SD Card:

- Download the latest stable files from [UUU GitHub page](#).
- //Linux
 - `$ chmod a+x uuu`
 - `$ sudo ./uuu -b sd_all imx-image-full-imx93frdm.rootfs.wic.zst`
- //Windows
 - `uuu.exe -b sd_all imx-image-full-imx93frdm.rootfs.wic.zst`

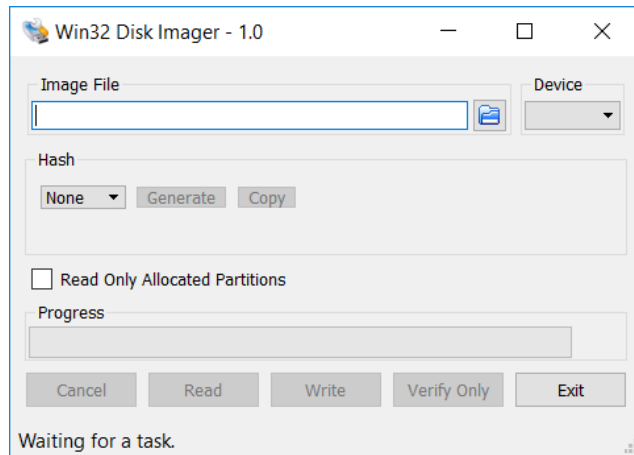
OR

2. Flashing an SD card image by linux

- `$ zstdcat imx-image-full-imx93frdm.rootfs.wic.zst | sudo dd of=/dev/sdx bs=1M && sync`

Flashing an SD card image by Windows Win32Diskmanager

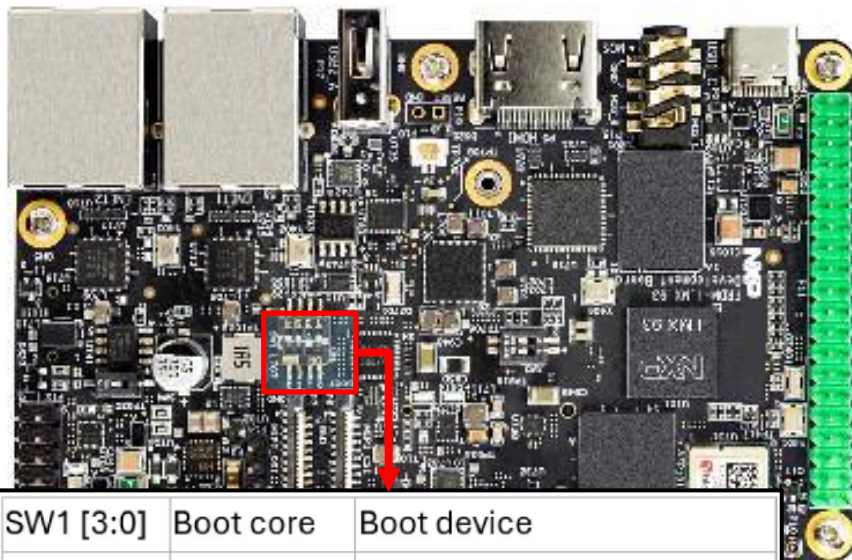
<https://win32diskimager.org/>



/Getting Started with FRDM i.MX93

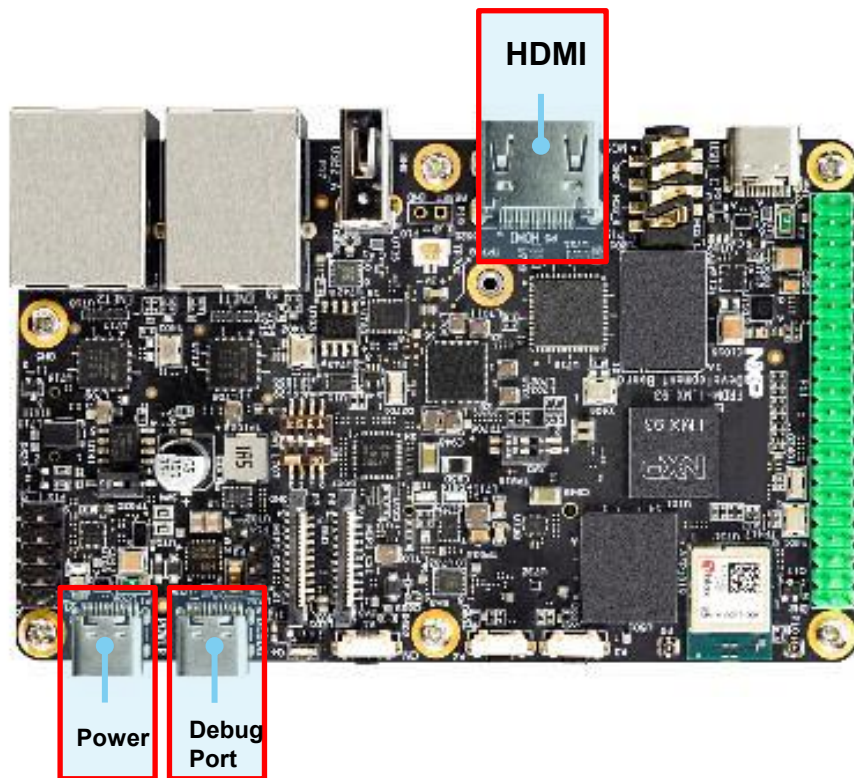
Step 3

Boot switch setup to 0011(SD boot)
or 0010(EMMC boot) recommend SD boot.



SW1 [3:0]	Boot core	Boot device
0001	Cortex-A55	Serial downloader (USB)
0010		uSDHC1 8-bit eMMC 5.1
0011		uSDHC2 4-bit SD3.0

/Getting Started with FRDM i.MX93



Step 4

Connect the power supply, HDMI cable, debug port and SD Card to the FRDM board.

After powering on, wait for a few seconds. If you see the penguin icon in the top left corner, the boot process is successful.



/ FRDM-IMX93 SW Release Package

Yocto Release

- <https://github.com/nxp-imx/imx-manifest>

Debian Release

- <https://github.com/nxp/flexbuild>

/ FRDM-IMX93 Yocto Release - BSP

Based on i.MX SW 2024 Q3 release

- Linux kernel: 6.6.36
- u-boot: 2024.04

Source: <https://github.com/nxp-imx-support/meta-imx-frdm>

FRDM-IMX93 BSP changes:

- U-boot:
 - Add basic support for FRDM-IMX93;
- Kernel:
 - Add basic support for FRDM-IMX93;
 - Add support for kinds of accessories;
- GoPoint:
 - Add FRDM-IMX93 support;
- FRDM-IMX93 Yocto layer:
 - Add Yocto layer for FRDM-IMX93;
 - Integrate u-boot/kernel/GoPoint patches;

```

├── layer.conf
├── machine
│   ├── imx93-11x11-lpddr4x-frdm.conf
│   └── imx93frdm.conf
├── recipes-bsp
│   ├── u-boot
│   │   ├── u-boot-imx
│   │   │   ├── 0001-net-phy-motorcomm-Add-support-for-YT8521-PHY.patch
│   │   │   ├── 0002-imx-imx93_frdr-Add-basic-board-support.patch
│   │   │   └── u-boot-imx_2024.04.bbappend
│   └── recipes-kernel
│       ├── linux
│       │   ├── linux-imx
│       │   │   ├── 0001-gpio-pca953x-fix-pca953x_irq_bus_sync_unlock-race.patch
│       │   │   ├── 0002-arm64-dts-add-i.MX93-11x11-FRDM-basic-support.patch
│       │   │   ├── 0003-arm64-dts-add-imx93-11x11-frdm-mt9m114-dts.patch
│       │   │   ├── 0004-Add-DSI-Panel-for-imx93.patch
│       │   │   ├── 0005-Add-CTP-support-for-waveshare-panel.patch
│       │   │   ├── 0006-arm64-dts-add-imx93-11x11-frdm-tianma-wvga-panel-dts.patch
│       │   │   ├── 0007-arm64-dts-add-imx93-11x11-frdm-aud-hat-dts.patch
│       │   │   ├── 0008-arm64-dts-add-button-support.patch
│       │   │   ├── 0009-arm64-dts-add-imx93-11x11-frdm-ov5640-dts.patch
│       │   │   ├── 0010-arm64-dts-add-imx93-11x11-frdm-ld-dts-for-lpm.patch
│       │   │   ├── 0011-arm64-dts-add-pwm-function-of-the-LED.patch
│       │   │   ├── 0012-arm64-dts-add-imx93-11x11-frdm-8mic-dts.patch
│       │   │   └── 0013-arm64-dts-add-imx93-11x11-frdm-lpuart-dts.patch
│       │   └── linux-imx_6.6.bbappend
│       └── meta-nxp-demo-experience
│           ├── conf
│           │   └── layer.conf
│           ├── recipes-examples
│           │   ├── gopoint-base-apps
│           │   │   ├── files
│           │   │   │   ├── 0001-Add-support-for-i.MX93-FRDM-board.patch
│           │   │   │   └── gopoint-base-apps.bbappend
│           │   └── imx-ebike
│           │       ├── files
│           │       │   ├── 0001-ebike-vit-Add-support-for-imx93frdm-board.patch
│           │       │   └── imx-ebike.bbappend

```

/ FRDM-IMX93 Yocto Release Usage

Download i.MX Yocto Release:

- `$ mkdir imx-yocto-bsp`
- `$ cd imx-yocto-bsp`
- `$ repo init -u https://github.com/nxp-imx/imx-manifest -b imx-linux-scarthgap -m imx-6.6.36-2.1.0.xml`
- `$ repo sync`

Integrate i.MX FRDM layer into i.MX Yocto Project:

- `$: cd ./sources`
- `$: git clone https://github.com/nxp-imx-support/meta-imx-frdm.git`
- `$: cd meta-imx-frdm`
- `$: git checkout imx-frdm-4.0`

Yocto Project Setup:

- `$ MACHINE=imx93frdm DISTRO=fsl-imx-xwayland source sources/meta-imx-frdm/tools/imx-frdm-setup.sh -b frdm-imx93`

Build images:

- `$ bitbake imx-image-full`

Flashing an SD card image:

- `$ zstdcat imx-image-full-imx93frdm.rootfs.wic.zst | sudo dd of=/dev/sdb bs=1M && sync`

Using uuu to burn image and rootfs to SD:

- `$ uuu -b sd_all imx-boot-imx93frdm-sd.bin-flash_singleboot <rootfs.wic.zst>`

/ FRDM-MX93 Debian Release

Debian is a free Operating System (OS), also known as Debian GNU/Linux.

i.MX Debian Linux SDK distribution is a combination of NXP-provided kernel and boot loaders with a Debian distro user-space image.

- Debian 12
- NXP packages are based i.MX SW Release 2024 Q3

i.MX Debian Linux SDK distribution uses Flexbuild to build system.

- Debian-based RootFS;
 - Debian Base (basic packages)
 - Debian Server (more packages without GUI Desktop)
 - Debian Desktop (with GNOME GUI Desktop)
- Linux kernel;
- BSP components;
- various applications (graphics, multimedia, networking, connectivity, security, and AI/ML);

Source: <https://github.com/NXP/flexbuild>

Introduction: <https://nxp.com/nxpdebian>

/ Quick Start with Debian

Flexbuild compiles and assembles the distro images as three parts:

- BSP firmware image
- Boot image
- RootFS image

Creating an SD card on the Linux host

- Download flex-installer
 - `$ wget http://www.nxp.com/lgfiles/sdk/lSDK2512/flex-installer`
 - `$ chmod +x flex-installer; sudo mv flex-installer /usr/bin`
- Plug the SD card into the Linux host and install the images as below:
 - `$ flex-installer -i pf -d /dev/sdb` (format SD card)
 - `$ flex-installer -i auto -d /dev/mmcblk1 -m imx93frdm` (automatically download and install images)
- Plug the SD card into the i.MX board and install the extra packages as follows:
 - `$ dhclient -i end0` (setup Ethernet network interface by DHCP or setting it manually)
 - `$ date -s "28 Mar 2026 09:00:00"` (setting correct system time is required)
 - `$ debian-post-install-pkg desktop` (install extra packages for GNOME GUI Desktop version)
 - or
 - `$ debian-post-install-pkg server` (install extra packages for Server version without GUI Desktop)
 - # After finishing the installation, run the reboot command to boot up the Debian Desktop/Server system.

/ Building Debain with FRDM i.MX93

(1) Download flex-installer :

```
$ wget http://www.nxp.com/lgfiles/sdk/lsdk2512/flex-installer
$ chmod +x flex-installer; sudo mv flex-installer /usr/bin
```

(2) Download FlexBuild Release:

```
$ git clone https://github.com/nxp/flexbuild
```

(3) Install Docker

```
$ sudo apt install docker.io
```

(4) Building Debian images in Flexbuild:

```
$ cd flexbuild
$ . setup.env (in host environment)
$ bld docker (create or attach to docker)
$ . setup.env (in docker environment)
$ bld host-dep (install host dependent packages)
$ bld -m imx93frdm
```

(4) Burn Image to SD Card:

```
$ cd build_lsdk2512/images
$ flex-installer -i pf -d /dev/mmcblkX
$ flex-installer -d /dev/mmcblk1 -m imx93frdm \
    -f firmware_imx93frdm_sdboot.img \
    -b boot_IMX_arm64_lts_6.12.20 \
    -r rootfs_lsdk2512_debian_imx93frdm.tar.zst
```

/ Building Debian Images with Flexbuild usage

Flexbuild usage:

- `$ bld -m imx93frdm` (build all images for imx93frdm)
- `$ bld uboot -m imx93frdm` (compile u-boot image for imx93frdm)
- `$ bld linux` (compile linux kernel for all arm64 i.MX machines)
- `$ bld bsp -m imx93frdm` (generate BSP firmware)
- `$ bld boot` (generate boot partition tarball including kernel, dtb, modules, distro bootscript for iMX machines)
- `$ bld multimedia` (build multimedia components for i.MX platforms)
- `$ bld rfs -r debian:base` (generate Debian base rootfs with base packages)
- `$ bld apps -r debian:server` (compile apps against runtime dependencies of Debian server RootFS)
- `$ bld merge-apps -r debian:server` (merge iMX-specific apps into target Debian server RootFS)
- `$ bld packrfs` (pack and compress target rootfs)



/ Related Documentation



FRDM-IMX93 Documents:

FRDM-IMX93 Quick Start Guide
FRDM-IMX93 Board User Manual
FRDM-IMX93 Software User Guide



More information about i.MX
productions can be found
at(<http://www.nxp.com/imxlinux>)

Debug Terminal

/ Debug Terminal in Windows

Serial Communication Console Setup

The WCH USB-serial chip on FRDM-IMX93 enumerates 2 serial ports. Assume that the ports are **COM11**, COM12. The first Port(**COM11**) is for the serial console communication from Arm A55 core. The serial-to-USB drivers are available at **CH342F Windows Driver** .

https://www.wch-ic.com/downloads/CH343SER_EXE.html

Note: To determine the port number of the i.MX board virtual COM port, open the Windows device manager and find USB serial Port in Ports (COM and LPT).

Recommend terminal-emulation Windows application: **Tera Term, PuTTY**

Serial port settings to 115,200 baud rate, 8 data bits, no parity and 1 stop bit.

/ Debug Terminal in Linux

Serial Communication Console Setup

The serial-to-USB drivers are available at **CH342F Linux Drivers** .

https://github.com/WCHSoftGroup/ch343ser_linux

On the command prompt of the Linux host machine, run the following command to determine the port number:

```
ls /dev/ttyCH343USB*
```

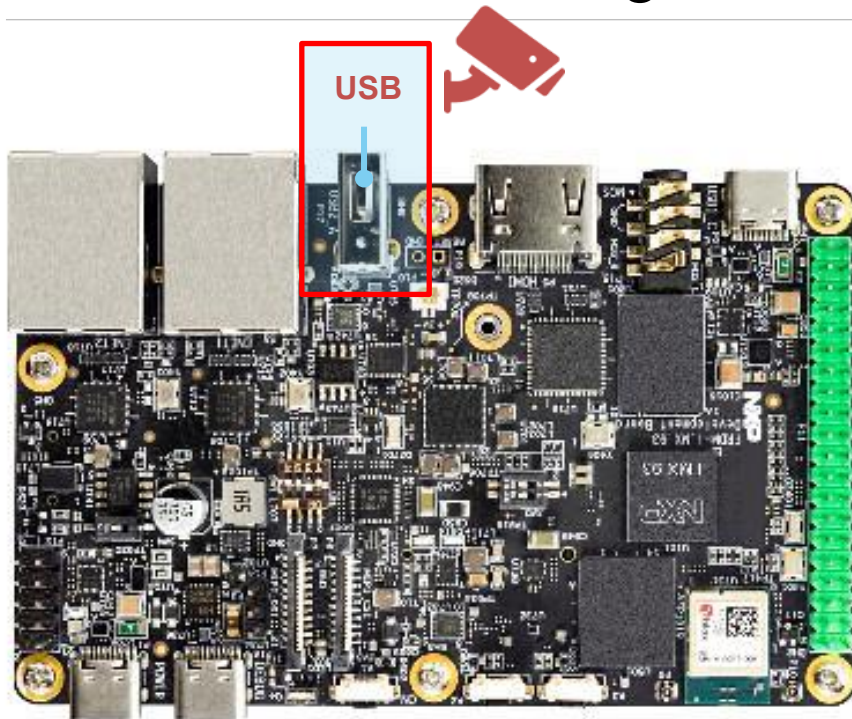
The first number is for Arm A55 core.

Recommend terminal-emulation Windows application: **Minicom**.

Serial port settings to 115,200 baud rate, 8 data bits, no parity and 1 stop bit.

Working with accessories

Camera Streaming with FRDM i.MX93



Using GStreamer to open USB Camera.

1. Connect USB Camera to FRDM board.
2. Confirm Camera Device

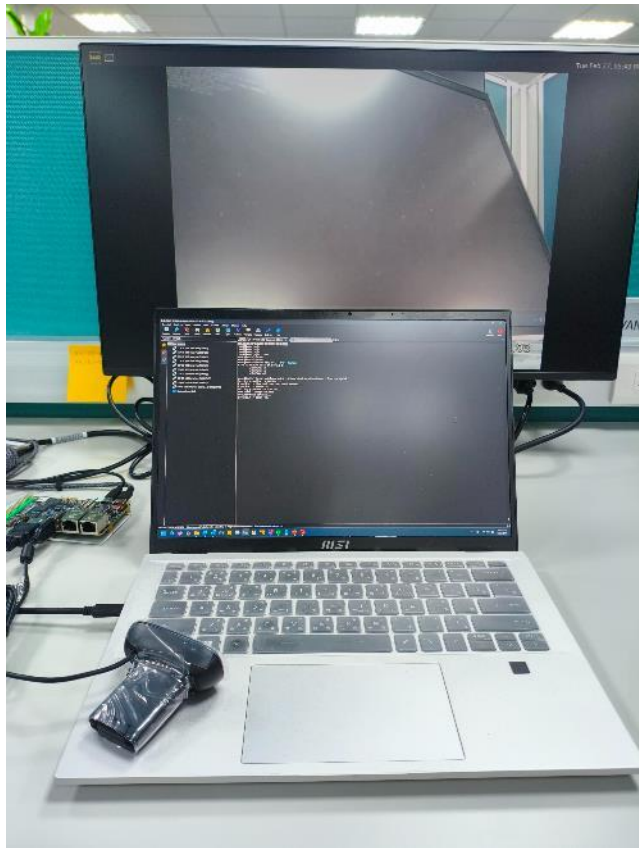
```
root@imx93frdm:~# v4l2-ctl --list-devices
C270 HD WEBCAM (usb-ci_hdrc.1-1):
    /dev/video0
    /dev/video1
    /dev/media0
```

3. Running Camera

```
root@imx93frdm:~# gst-launch-1.0 v4l2src
device=/dev/video0 ! fpsdisplaysink
```

```
Setting pipeline to PAUSED ...
Pipeline is live and does not need PREROLL ...
Pipeline is PREROLLED ...
Setting pipeline to PLAYING ...
New clock: GstSystemClock
Redistribute latency...
Total showed frames (3401), playing for
(0:01:57.471764157), fps (28.952).
```

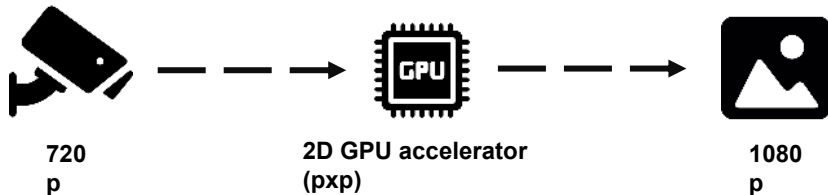
/ Display camera video with gst-launch



```
root@imx93frdm:~# v4l2-ctl --list-devices
C270 HD WEBCAM (usb-ci_hdrc.1-1):
    /dev/video0
    /dev/video1
    /dev/media0

root@imx93frdm:~# gst-launch-1.0 v4l2src device=/dev/video0 !
fpsdisplaysink
Setting pipeline to PAUSED ...
Pipeline is live and does not need PREROLL ...
Pipeline is PREROLLED ...
Setting pipeline to PLAYING ...
New clock: GstSystemClock
Redistribute latency...
^Chandling interrupt.
Interrupt: Stopping pipeline ...
Execution ended after 0:04:00.767927698
Setting pipeline to NULL ...
Total showed frames (6333), playing for (0:04:00.767904573), fps
(26.303).
Freeing pipeline ...
```

/ Camera Streaming with 2D GPU



Utilize GStreamer in combination with the GPU to accelerate video processing.

```
root@imx93frdm:~# gst-launch-1.0 v4l2src device=/dev/video0 !  
image/jpeg, width=1280, height=720, framerate=30/1 !  
jpegdec ! queue ! imxvideoconvert_pxp ! video/x-raw,  
width=1920, height=1080 ! waylandsink
```

Setting pipeline to PAUSED ...

Pipeline is live and does not need PREROLL ...

Pipeline is PREROLLED ...

Setting pipeline to PLAYING ...

New clock: GstSystemClock

Redistribute latency...

Total showed frames (138), playing for (0:00:11.492181461), fps (12.008).

Bring up Wi-Fi interfaces

1. Power ON the board in the internal boot mode and wait until the board is completely booted.
2. To login into the Linux running on the FRDM i.MX 93, use “**root**” as the default login username for the i.MX Linux OS. There is no password.

```
NXP i.MX Release Distro 6.6-scarthgap imx93frdm-iwxxx-matter
ttyLP0

imx93frdm-iwxxx-matterlogin: root
root@imx93frdm-iwxxx-matter:~#
```

Bring up Wi-Fi interfaces

3. Verify the module parameter configuration file with below given configuration using the nano editor tool.

```
nano /lib/firmware/nxp/wifi_mod_para.conf
```

```
SDIW612 = {  
    cfg80211_wext=0xf  
    max_vir_bss=1  
    cal_data_cfg=none  
    ps_mode=1  
    auto_ds=1  
    host_mlme=1  
    fw_name=nxp/sduart_nw61x_v1.bin.se  
}
```

4. Load the drivers and firmware.

```
modprobe moal mod_para=nxp/wifi_mod_para.conf
```

Bring up Wi-Fi interfaces

5. Verify that the firmware was downloaded successfully via driver logs on the console.

`dmesg -w`

```
[ 59.848148] wlan: loading out-of-tree module taints kernel.
[ 59.870831] wlan: Loading MWLAN driver
[ 59.871185] wlan: Register to Bus Driver...
[ 59.871381] vendor=0x0471 device=0x0205 class=0 function=1
[ 59.871472] Attach moal handle ops, card interface type: 0x109
[ 59.871483] rps set to 0 from module param
[ 59.872310] SDIW612: init module param from usr cfg
[ 59.872376] card_type: SDIW612, config block: 0
[ 59.872387] cfg80211 wext=0xf
[ 59.872390] max_vir_bss=1
[ 59.872395] cal_data_cfg=none
[ 59.872398] ps_mode = 1
[ 59.872400] auto_ds = 1
[ 59.872407] host_mlme=enable
[ 59.872411] fw_name=nxp/sduart_nw61x_v1.bin.se
[ 59.872432] SDIO: max_segs=128 max_seg_size=65535
[ 59.872437] rx_work=1 cpu num=2
[ 59.872442] Enable moal_recv_amsdu_packet
[ 59.872461] Attach wlan adapter operations.card type is 0x109.
[ 59.872745] wlan: Enable TX SG mode
[ 59.872754] wlan: Enable RX SG mode
```

```
[ 59.872745] wlan: Enable TX SG mode
[ 59.872754] wlan: Enable RX SG mode
[ 59.880985] Request firmware: nxp/sduart_nw61x_v1.bin.se
[ 60.160989] Wlan: FW download over, firmwarelen=1065364
downloaded 953164
[ 60.588889] WLAN FW is active
[ 60.588903] on time is 60412073569
[ 60.615581] VDLL image: len=112200
[ 60.616635] fw_cap_info=0x487cff03, dev_cap_mask=0xffffffff
[ 60.616654] uuid: 544906e672fd5192ac42d748be0e5109
[ 60.616659] max_p2p_conn = 8, max_sta_conn = 16
[ 60.738506] Register NXP 802.11 Adapter wlan0
[ 60.738599] wlan: uaptd set max_mtu 2000
[ 60.744185] Register NXP 802.11 Adapter uap0
[ 60.838830] Register NXP 802.11 Adapter wfd0
[ 60.838874] wlan: version = SDIW612---18.99.3.p21.10-
MM6X18505.p4-GPL-(FP92)
[ 60.868735] wlan: Register to Bus Driver Done
[ 60.868751] wlan: Driver loaded successfully
```

Bring up Wi-Fi interfaces

6. Once the driver modules loaded successfully, verify the Wi-Fi interface created by the driver using below given command.

`ifconfig -a`

```
mlan0: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 20:ba:36:5c:e6:c9 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0

uap0: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 22:ba:36:5c:e7:c9 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0

wfd0: flags=4098<BROADCAST,MULTICAST> mtu 1500
    ether 22:ba:36:5c:e6:c9 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions
0
```


Create STA Mode and connect to Ext AP

This section describes configuration for the STA mode on the FRDM i.MX 93 board using the **/etc/wpa_supplicant.conf** configuration file.

The wpa_supplicant utility is configured using a text file that lists all expected networks and security policies, including pre-shared keys.

1. Open configuration file /etc/wpa_supplicant.conf which includes the network details of the Ext-AP (Here Ext-AP could be your home/office Wi-Fi router or Mobile hotspot) that you want to connect to.

Sample wpa_supplicant.conf with WPA2 security:

```
$ nano /etc/wpa_supplicant.conf
```

```
ctrl_interface=/var/run/wpa_supplicant
ctrl_interface_group=0
update_config=1
network={
    ssid="NXP_Demo" #modify
    key_mgmt=WPA-PSK
    psk="123456789" #modify
}
```

Create STA Mode and connect to Ext AP

2. Bring up the wlan0 interface and verify the STA mode interface using following command.

```
ifconfig wlan0 up  
ifconfig wlan0
```

```
wlan0: flags=4098<BROADCAST,MULTICAST> mtu 1500  
    ether 20:ba:36:5c:e6:c9 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions  
0
```

Create STA Mode and connect to Ext AP

3. Make sure to stop any running instance of the wpa_supplicant to avoid failures.

```
killall wpa_supplicant
```

```
root@imx93frdm-iwxxx-matter:~# killall wpa_supplicant
root@imx93frdm-iwxxx-matter:~# killall wpa_supplicant
wpa_supplicant: no process found
```

Create STA Mode and connect to Ext AP

4. Use following command to start wpa_supplicant in background.

```
$ wpa_supplicant -imlan0 -Dnl80211 -c /etc/wpa_supplicant.conf -B
```

```
root@imx93frdm-iwxxx-matter:~# wpa_supplicant -imlan0 -Dnl80211 -  
c /etc/wpa_supplicant.conf -B  
Successfully initialized wpa_supplicant  
rfkill: Cannot open RFKILL control device  
root@imx93frdm-iwxxx-matter:~# [ 1278.242273] wlan: mlan0 START  
SCAN  
[ 1283.505491] wlan: SCAN COMPLETED: scanned AP count=51  
[ 1283.519298] wlan: HostMlme mlan0 send auth to bssid  
be:XX:XX:XX:b0:40  
[ 1283.522723] mlan0:  
[ 1283.522736] wlan: HostMlme Auth received from  
be:XX:XX:XX:b0:40 [ 1283.567080] wlan: HostMlme mlan0 Connected  
to bssid be:XX:XX:XX:b0:40 successfully  
[ 1283.582054] mlan0:  
[ 1283.582069] wlan: Send EAPOL pkt to be:XX:XX:XX:b0:40  
[ 1283.589059] mlan0:  
[ 1283.589072] wlan: Send EAPOL pkt to be:XX:XX:XX:b0:40  
[ 1283.596362] woal_cfg80211_set_rekey_data return:  
gtk_rekey_offload is DISABLE
```

Create STA Mode and connect to Ext AP

5. Once the connection is done successfully, start the udhcp client by using the following command

to get the dynamic IP address from the Ext-AP and it could be any IP address of the format “xxx.xxx.xxx.xxx”.

For example, in below snapshot the IP address “192.168.105.27” is obtained from the Ext-AP.

NOTE: Please wait for some time after executing the following command. On successful execution of the command it will return to root prompt by pressing “enter” key. If it does not return to root prompt, there is some failure.

```
$ udhcpc -i mlan0
```

```
root@imx93frdm-iwxxx-matter:~# udhcpc -i mlan0
udhcpc: started, v1.36.1
Dropped protocol specifier '.udhcpc' from 'mlan0.udhcpc'. Using
'mlan0' (ifindex=4).
udhcpc: broadcasting discover
udhcpc: broadcasting select for 192.168.105.27, server
192.168.105.68
udhcpc: lease of 192.168.105.27 obtained from 192.168.105.68,
lease time 3599
/etc/udhcpc.d/50default: Adding DNS 192.168.105.68
Dropped protocol specifier '.udhcpc' from 'mlan0.udhcpc'. Using
'mlan0' (ifindex=4).
root@imx93frdm-iwxxx-matter:~#
```

Create STA Mode and connect to Ext AP

6. Verify IP address assigned to STA device interface using following command.
For example, in below snapshot the IP address “192.168.105.27” is obtained from the Ext-AP.

```
$ ifconfig wlan0  
$ ping 8.8.8.8  
$ ping www.google.com
```

```
root@imx93frdm-iwxxx-matter:~# ifconfig wlan0  
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
        inet 192.168.105.27 netmask 255.255.255.0 broadcast  
192.168.105.255  
        inet6 fe80::22ba:36ff:fe5c:e6c9 prefixlen 64 scopeid  
0x20<link>  
        ether 20:ba:36:5c:e6:c9 txqueuelen 1000 (Ethernet)  
RX packets 7 bytes 1212 (1.1 KiB)  
RX errors 0 dropped 0 overruns 0 frame 0  
TX packets 36 bytes 5328 (5.2 KiB)  
TX errors 0 dropped 0 overruns 0 carrier 0 collisions  
0  
root@imx93frdm-iwxxx-matter:~#
```

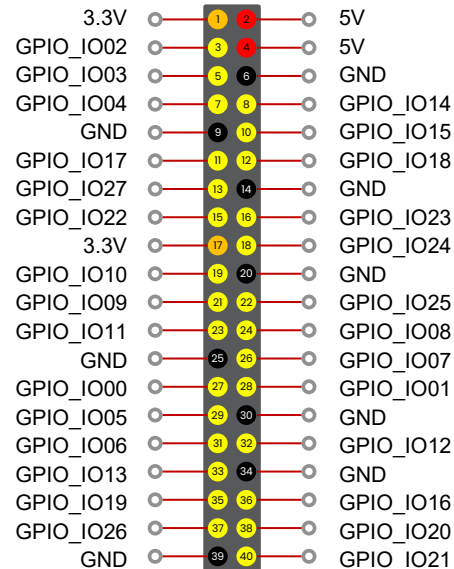
/ Basic IO control

Config GPIO

```
root@imx93frdm:/# cat /sys/kernel/debug/pinctrl/*/pins
registered pins: 108
pin 0 (IMX93_IOMUXC_DAP_TDI) 28:43820000.gpio 443c0000.pinctrl
pin 1 (IMX93_IOMUXC_DAP_TMS_SWDIO) 29:43820000.gpio 443c0000.pinctrl
pin 2 (IMX93_IOMUXC_DAP_TCLK_SWCLK) 30:43820000.gpio 443c0000.pinctrl
pin 3 (IMX93_IOMUXC_DAP_TDO_TRACESWO) 31:43820000.gpio 443c0000.pinctrl
pin 4 (IMX93_IOMUXC_GPIO_I000) 0:43810000.gpio 443c0000.pinctrl
pin 5 (IMX93_IOMUXC_GPIO_I001) 1:43810000.gpio 443c0000.pinctrl
pin 6 (IMX93_IOMUXC_GPIO_I002) 2:43810000.gpio 443c0000.pinctrl
pin 7 (IMX93_IOMUXC_GPIO_I003) 3:43810000.gpio 443c0000.pinctrl
```

```
root@imx93frdm:/# gpiodetect
gpiochip0 [43810000.gpio] (32 lines)
gpiochip1 [43820000.gpio] (32 lines)
gpiochip2 [43830000.gpio] (32 lines)
gpiochip3 [47400000.gpio] (32 lines)
gpiochip4 [1-0022] (24 lines)
gpiochip5 [0-0020] (8 lines)
```

```
root@imx93frdm:/# gpioset -i -c gpiochip0 2=1
gpioset> set 2=1
gpioset> get 2
"2"=active
gpioset> set 2=0
gpioset> get 2
"2"=inactive
gpioset>
gpioset> exit
root@imx93frdm:/#
```



Yellow circle: GPIO Black circle: Ground Yellow circle: 3.3V Red circle: 5V

How to upgrade Linux Kernel and dtb on eMMC without UUU

NPU introduction

/ NPU with FRDM i.MX93

- ✓ **Using arm Ethos-U65 of NPU**
(<https://developer.arm.com/Processors/Ethos-U65>)
- ✓ **Higher-Efficiency Performance : 0.5 TOP/s @ 1 GHz**
- ✓ **Support 8-bit integer weight operation.**
- ✓ **Support 8-bit and 16-bit integer activations operators.**
- ✓ **Support ML Framework at MPU Chip, including TensorFlow Lite.**
(NXP i.MX8M PLUS / i.MX93 Series)
- ✓ **Non-need Warm-Up Process**

NPU with FRDM i.MX93 – Model Converter

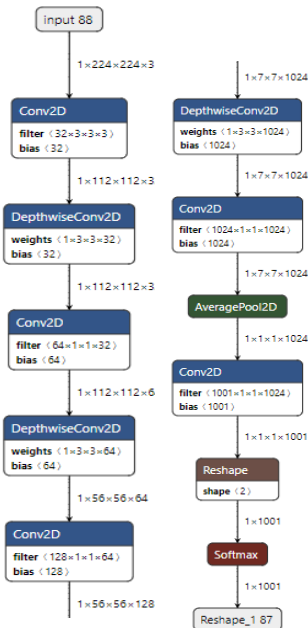


Quantized
Model
(*.tflite)

Vela Compile Tool



Vela Model
(*-vela.tflite)

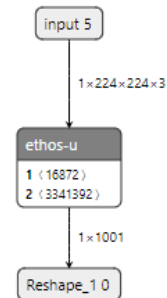


General Method :
\$ vela model.tflite

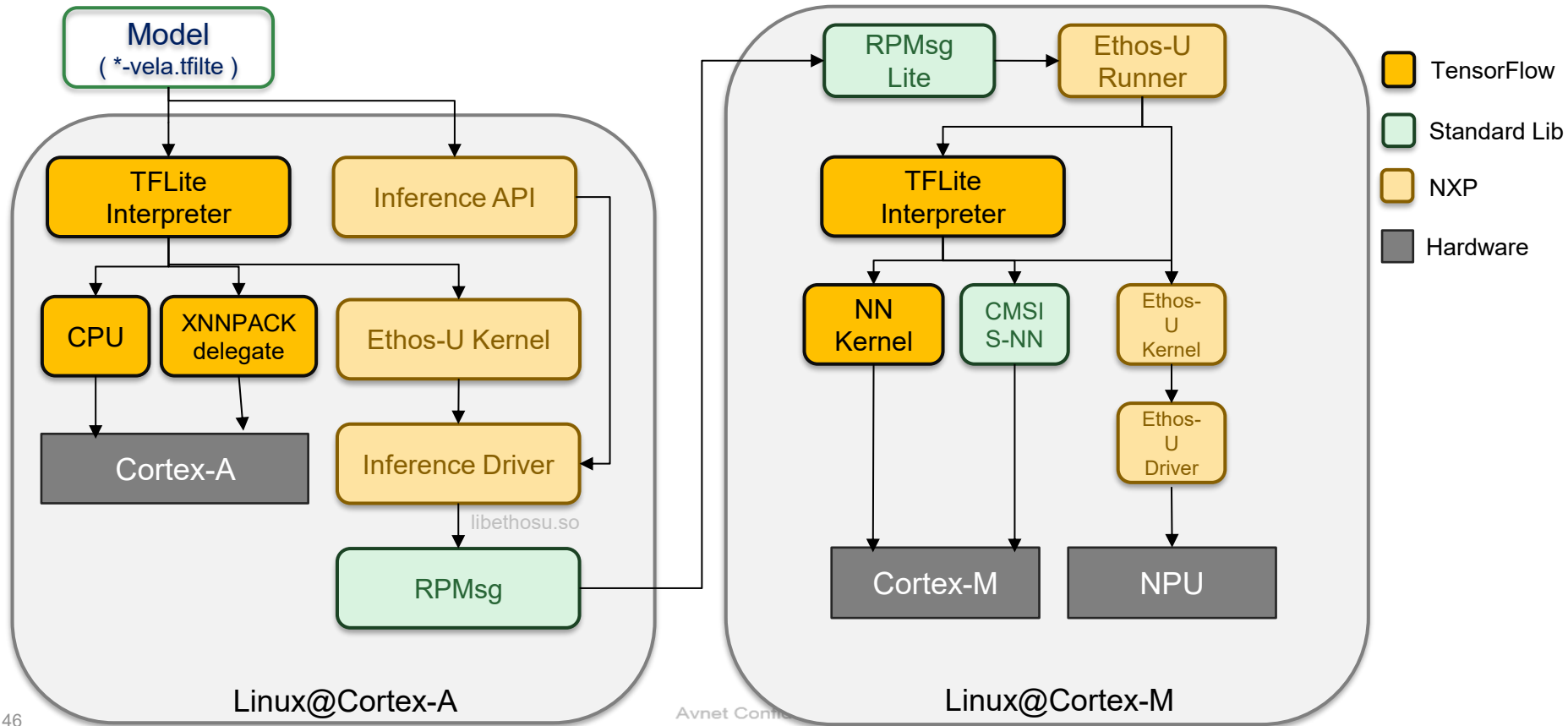
Highly-Performance Method :
\$ vela --optimise Performance
model.tflite

Minimum-Size Method :
\$ vela --optimise Size model.tflite

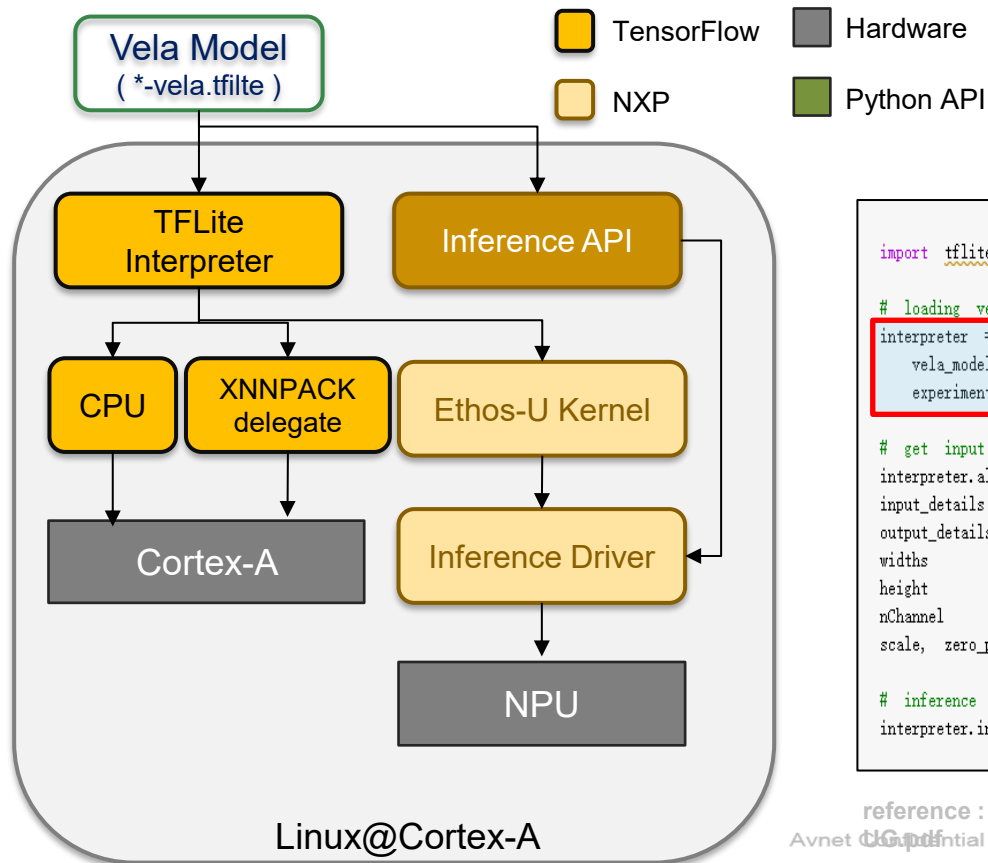
Help :
\$ vela --help



/ NPU with FRDM i.MX93



/NPU with FRDM i.MX93



How to use NPU by Python

```
import tflite_runtime.interpreter as tflite

# loading vela model
interpreter = tflite.Interpreter(
    vela_model,
    experimental_delegates=[tflite.load_delegate("/usr/lib/libethosu_delegate.so")])

# get input and output information
interpreter.allocate_tensors()
input_details = interpreter.get_input_details()
output_details = interpreter.get_output_details()
widths = input_details[0]['shape'][2]
height = input_details[0]['shape'][1]
nChannel = input_details[0]['shape'][3]
scale, zero_point = input_details[0]['quantization']

# inference
interpreter.invoke()
```

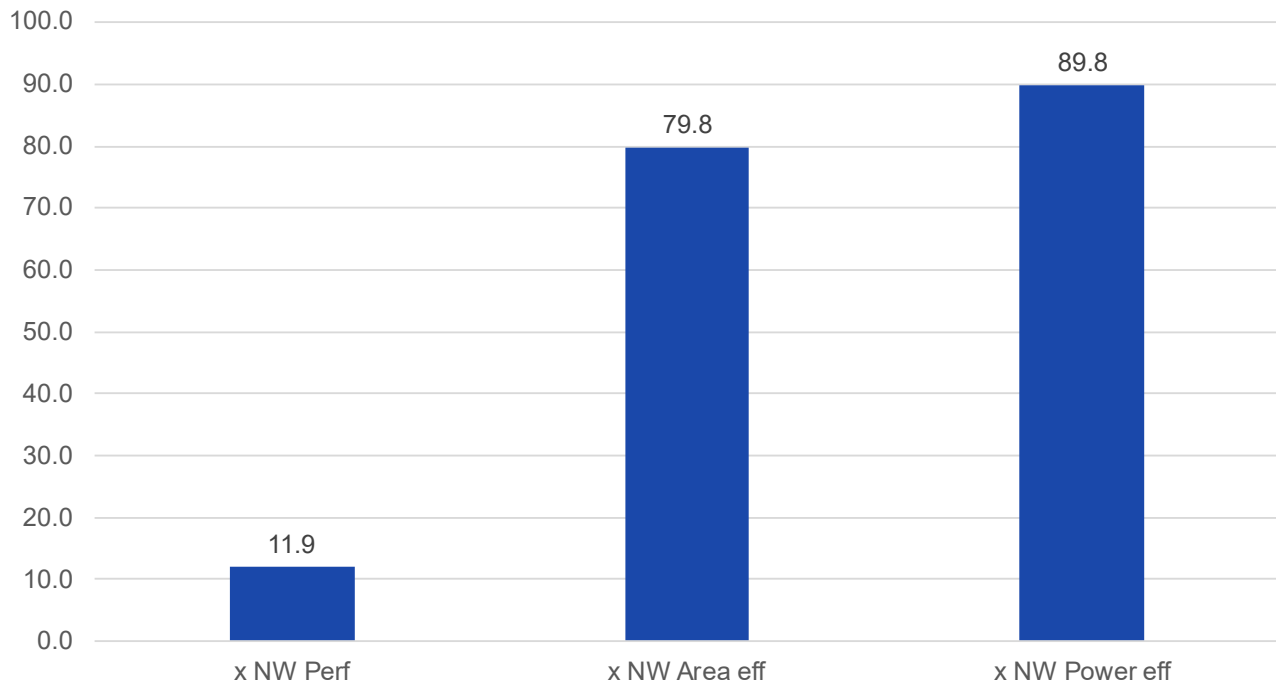
See Here

reference : <https://www.nxp.com/docs/en/user-guide/IMX-MACHINE-LEARNING-UG.pdf>

/ NPU with FRDM i.MX93 - Performance

i.MX93 (Ethos-U65) vs Cortex-A53

x Times Cortex-A53 MP4 @ 1.8Ghz



With Ethos-U65, we get

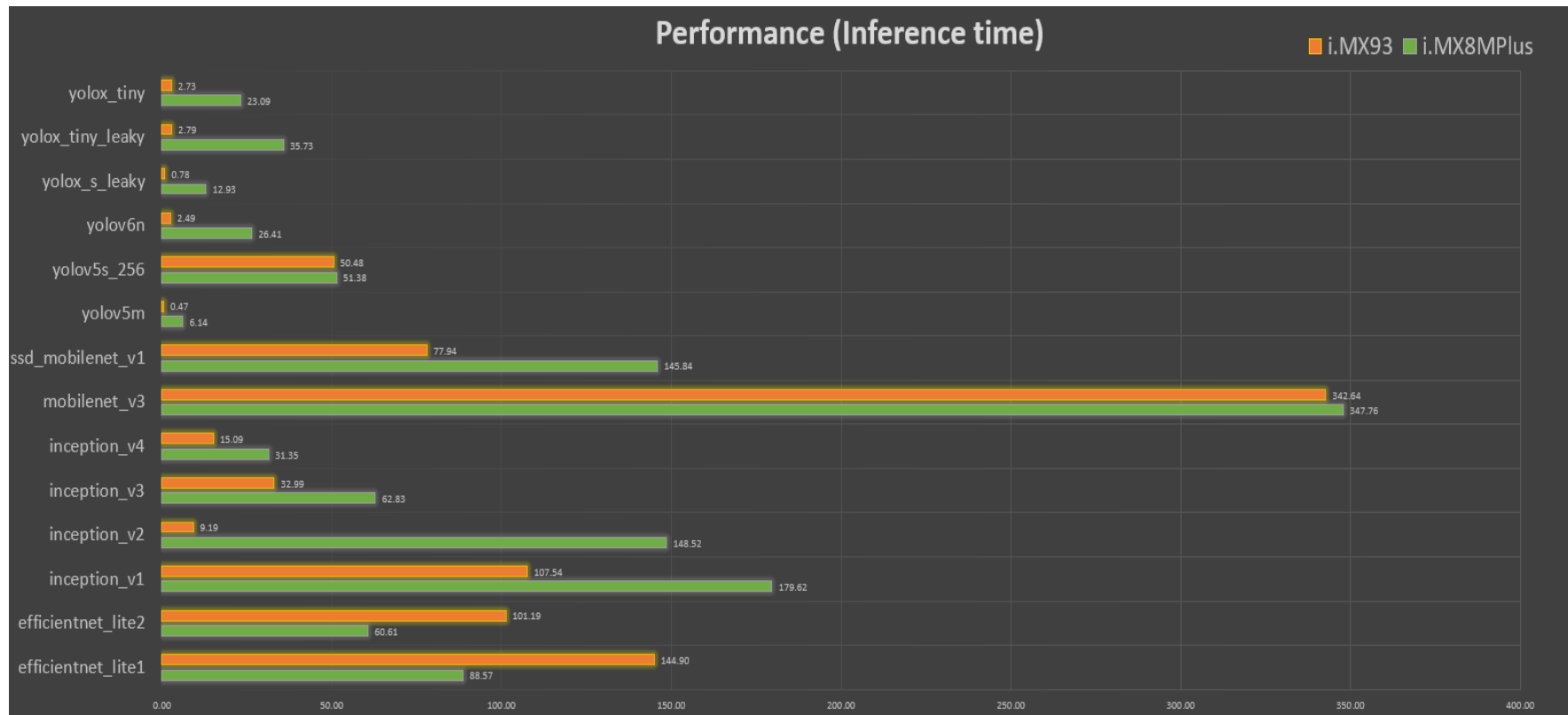
11.9x times performance

89.8x times power efficiency

Over Cortex-A53

/ NPU with FRDM i.MX93 - Performance

i.MX93 (Ethos-U65) vs i.MX 8M Plus



/ FRDM i.MX93 Target Application

Industrial Automation	Smart Home	Smart City	Automotive
- Industrial Machine vision - Industrial scanning/printing	- Smart doorbell - Smart lock - Smart home hub/ Hue Bridge	- Smart lighting - Traffic control	- Driver Monitoring System (DMS) - Object Monitoring System (OMS)
			

Industrial Automation and Building Control



Industrial Automation	Building Control & Energy
- Industrial HMI - Industrial gateway - I/O control - Combines with i.MX RT1180 crossover MCU for larger gateways and IO controller	- Access control - Energy meter - Energy grid equipment - EV charging station - Environmental control - Sensor hub



/ Useful Link with FRDM i.MX93

NXP offer

- Getting Started with FRDM-IMX93 <https://www.nxp.com/document/guide/getting-started-with-frdm-imx93:GS-FRDM-IMX93>
- i.MX FRDM Software User Guide <https://docs.nxp.com/bundle/UG10195/page/topics/introduction.html>
- **Debian** documents at <http://www.nxp.com/nxpdebian>
- GoPoint repo: <https://github.com/nxp-imx-support/nxp-demo-experience-demos-list>
- Image Classification/Object Detection: <https://github.com/nxp-imx/eiq-example>
- i.MX Smart Fitness: <https://github.com/nxp-imx-support/imx-smart-fitness>
- i.MX Smart Kitchen: <https://github.com/nxp-imx-support/smart-kitchen>
- i.MX E-Bike VIT: <https://github.com/nxp-imx-support/imx-ebike-vit>
- eIQ toolkit : <https://www.nxp.com/design/design-center/software/eiq-ml-development-environment/eiq-toolkit-for-end-to-end-model-development-and-deployment:EIQ-TOOLKIT>

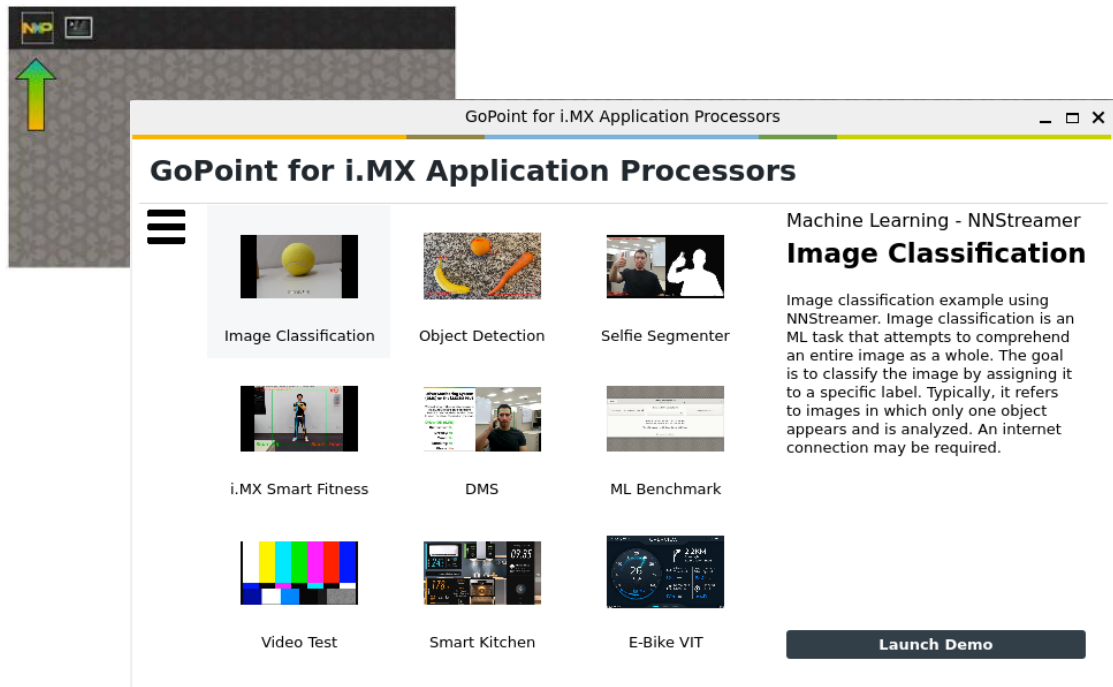
Thank you

Backup

GoPoint

/GoPoint Demo with FRDM i.MX93

Click on the NXP icon in the top left corner of the desktop to access the GoPoint Demo Menu

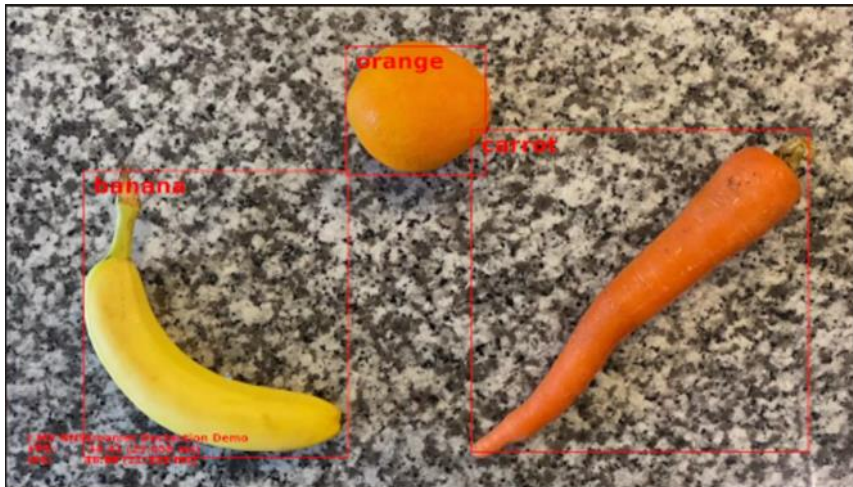


1. Image Classification
2. Object Detection
3. Selfie Segmenter
4. i.MX Smart Fitness
5. DMS (Driver Monitor System)
6. ML Benchmark
7. Video Test
8. i.MX Smart Kitchen
9. i.MX E-Bike VIT

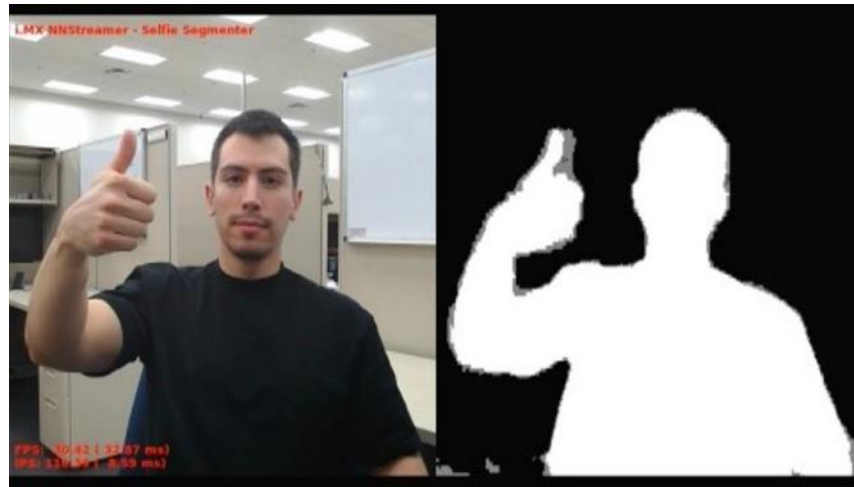
Please connect to the internet.

/GoPoint Demo with FRDM i.MX93

Object Detection

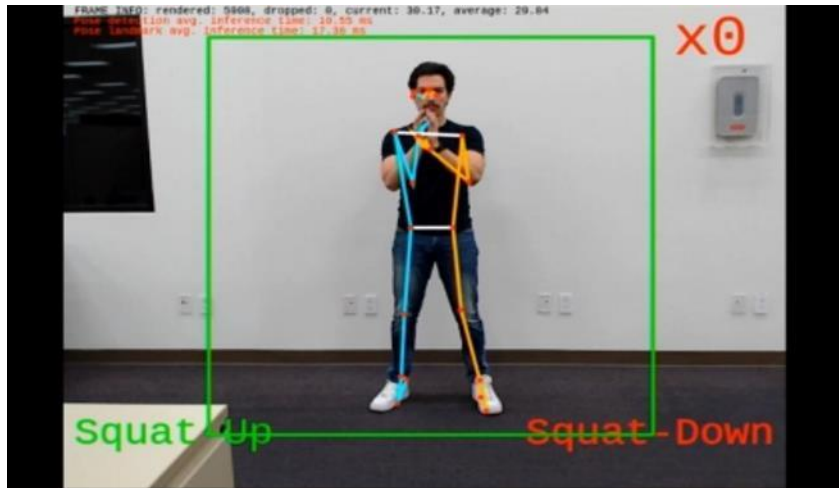


Selfie Segmenter



/GoPoint Demo with FRDM i.MX93

Smart Fitness



DMS

Driver Monitoring System (DMS) on the i.MX93

When driving, milliseconds can save a life. **i.MX93's on-board NPU** allows for quick and responsive alerts to keep the driver focused on the road.

Driver OK (2.0%)

Distracted: No

Drowsy: No

Yawn: No

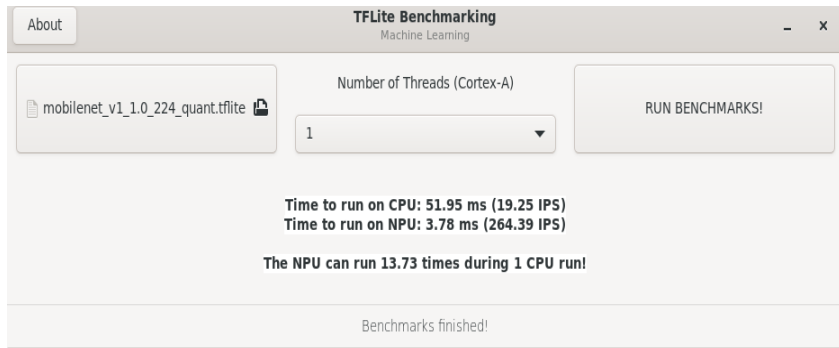
Smoking: No

Phone: Yes

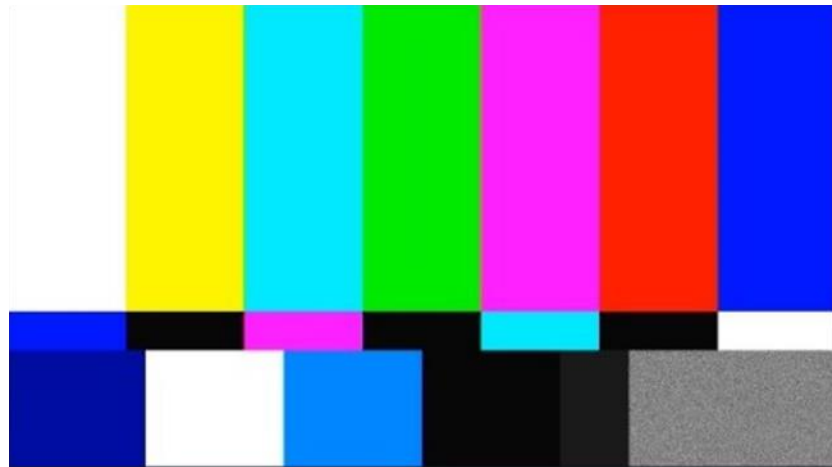


/GoPoint Demo with FRDM i.MX93

ML Benchmark



Video Test Demo



/ GoPoint Demo with FRDM i.MX93

Smart Kitchen



E-Bike VIT Demo

